

Algorithm: Acceptance–Rejection Sampling

1. Generate $Y \sim g$.
2. Generate $U \sim \text{Uniform}(0, 1)$, independent of Y .
3. If

$$U \leq \frac{f(Y)}{c g(Y)},$$

accept $X = Y$. Otherwise **reject** and go to Step 1.

- Note c must satisfy $c = \sup_x \frac{f(x)}{g(x)}$ (or any valid upper bound) for the ratio in Step 3 to always lie in $[0, 1]$.
- Works even when f is known only up to a normalizing constant – a property that becomes crucial later, in Bayesian computation (Module 5).

Why Does This Work?

Proof sketch

Condition on acceptance. The joint “proposal” density of (Y, U) is $g(y)$ on y times uniform on $[0, 1]$ for u . The accepted region is $\{(y, u) : 0 \leq u \leq f(y)/(c g(y))\}$.

$$\underbrace{P(Y \leq x \mid \text{accept})}_{\text{red underline}} = \frac{\int_{-\infty}^x g(y) \cdot \frac{f(y)}{c g(y)} dy}{\int_{-\infty}^{\infty} g(y) \cdot \frac{f(y)}{c g(y)} dy} = \frac{\int_{-\infty}^x f(y) dy}{\int_{-\infty}^{\infty} f(y) dy} = \overset{\text{X}}{\underbrace{F(x)}} \rightarrow f$$

So the *accepted* Y values have exactly the target CDF F . ■

Recall : $U \sim U(0,1)$

$Y \sim g$

$\downarrow$ Accept when $U \leq \frac{f(Y)}{c g(Y)} \rightarrow$

$$P(Y \leq x | \text{accept}) = \frac{P(Y \leq x, \text{accept})}{P(\text{accept})}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \rightarrow 1/c$$

$$P(Y \leq x, \text{accept}) = P\left(\underbrace{Y \leq x}_\wedge, U \leq \frac{f(Y)}{c g(Y)}\right)$$

$$= \int_{-\infty}^x P\left(U \leq \frac{f(y)}{c g(y)}\right) g(y) dy$$

$$P(Y \leq x) = \int_{-\infty}^x g(y) dy \quad \leftrightarrow$$

$$= \int_{-\infty}^x \left(\int_0^{\frac{f(y)}{c g(y)}} 1 du \right) g(y) dy \quad \begin{matrix} x \sim f \\ P(X \leq x) = F \end{matrix}$$

$$= \int_{-\infty}^x \frac{f(y)}{c g(y)} g(y) dy = \frac{1}{c} \int_{-\infty}^x f(y) dy$$

$$= \frac{1}{c} P(X \leq x)$$

$$= \frac{1}{c} F_X(x)$$

$(Y \sim g)$

$$P(Y \leq x | \text{accept}) = \frac{1/c F_X(x)}{1/c} = F_X(x) \rightarrow$$

$\rightarrow$ cdf of X

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- Each proposal Y is accepted with probability $P(\text{accept}) = \frac{1}{c}$. ✓

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- The number of proposals needed until one acceptance follows a Geometric distribution with mean c .

$\Rightarrow$

$$\begin{aligned} \text{Success prob} &= \frac{1}{c} \\ \Rightarrow \text{Geom}\left(\frac{1}{c}\right) \\ \Rightarrow \text{mean} &= c \end{aligned}$$

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Trade-off

A very tight envelope ($c \approx 1$) gives high efficiency but may be harder to construct or sample from; a loose, simple envelope is easy to sample but wastes many draws (small acceptance probability).

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Take **envelope** $g(x) = 1$ (i.e., Uniform(0, 1)), the simplest possible proposal.

$$c = \sup_{0 \leq x \leq 1} \frac{f(x)}{g(x)} = \sup_x 6x(1-x) = 6 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1.5 \text{ (at } x = 0.5\text{)}.$$

$$p(\text{accept}) = \frac{1}{1.5} \approx \frac{2}{3} = 0.66 \text{ } 66\%$$

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Algorithm

1. Generate $Y \sim \text{Unif}(0, 1)$.
2. Generate $U \sim \text{Unif}(0, 1)$.
3. Accept $X = Y$ if $U \leq \frac{6Y(1 - Y)}{1.5} = 4Y(1 - Y)$; else repeat.

Expected number of proposals per acceptance = $c = 1.5$ – quite efficient.



Example: Sampling from a Truncated Normal

Goal: sample $X \sim N(0, 1)$ truncated to $[a, \infty)$ for large $a > 0$ (relevant e.g. for rare-event / tail simulation).



$[X_1, X_2, \dots, X_n \sim \text{Normal} \rightarrow$

$(-\infty, \infty)$
 $\rightarrow$ accept-reject

want: simulate from $\text{Normal}(a, \infty)$
keep i if $X_i \geq a$

$$p(X > a) = \frac{0.01}{0.99}$$

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- Better proposal: a shifted **Exponential** density on $[a, \infty)$,

$$g(x) = a e^{-a(x-a)}, \quad x \geq a,$$

→ proper density

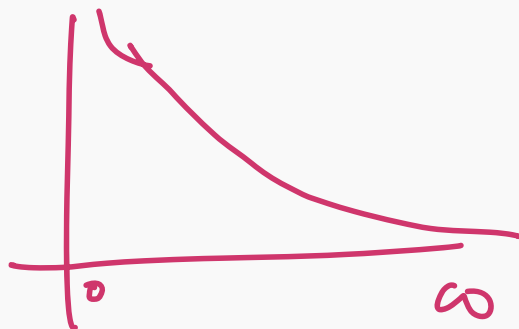
chosen because it decays at a similar rate to the Normal tail.

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$x > 0$

sub

$$C = \frac{f(x)}{g(x)}$$



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proposal

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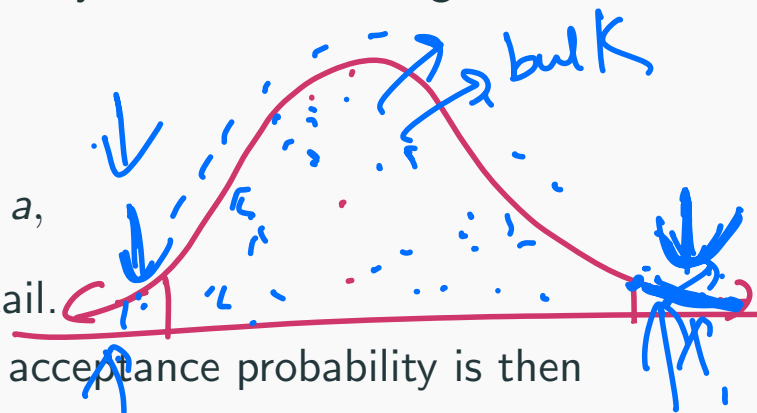
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Lesson

The choice of envelope g is where the “art” of acceptance–rejection lies – match tail behaviour to the target for high efficiency.

Outline

Pseudo-Random Number Generators

Generating Uniform and Non-Uniform Random Variables

Inverse Transform Method

Acceptance–Rejection Method

Box–Muller Transformation

Derivation

Basic Algorithm

Ratio-of-Uniforms Method

Diagnosing Randomness

Simulation Error

Unit Summary